

自动驾驶开发环境搭建

本实验将在Ubuntu系统中安装ROS，搭建可支持小车自动驾驶开发的基础环境。

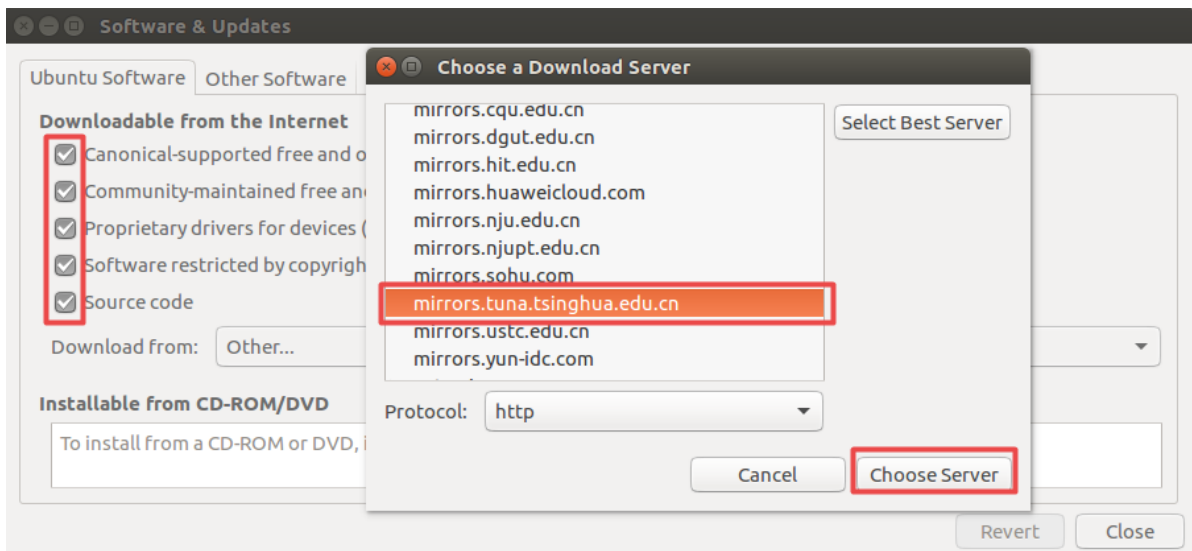
一 系统版本

- Ubuntu系统： 20.04
- ROS系统： ROS Kinetic

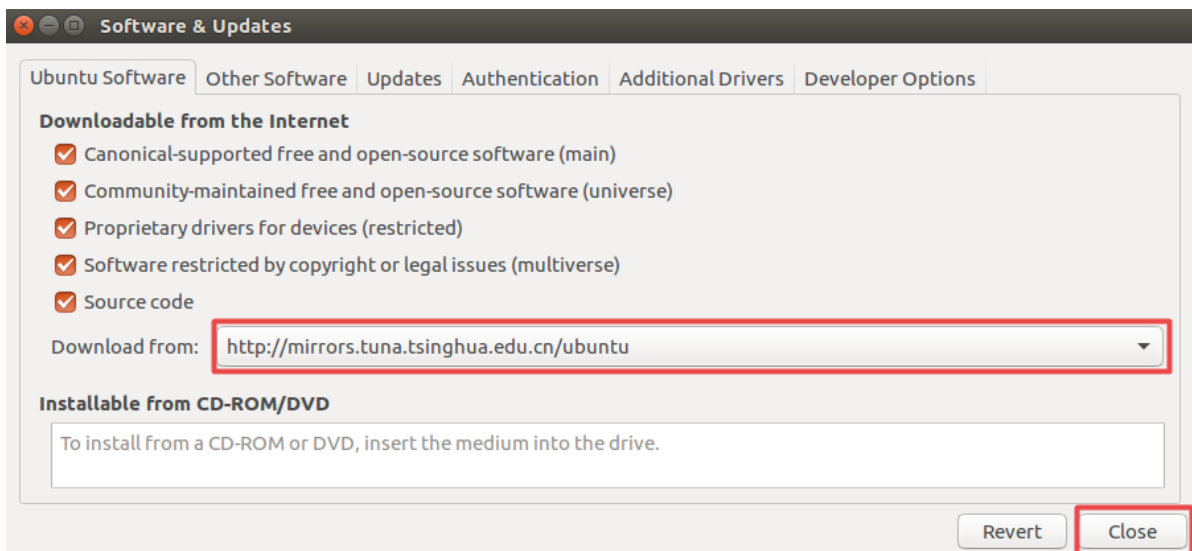
二 软件安装及配置

2.1 更改源，将下载源更改为清华源

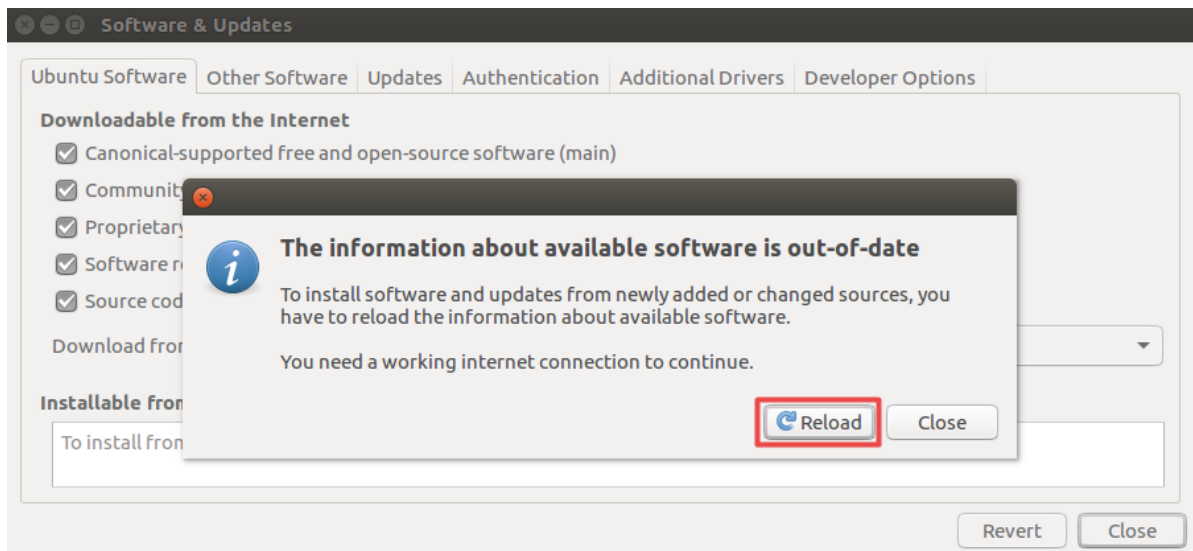
在Ubuntu设置--Software & Update中设置如下：



单击“选择服务器”后输入用户密码，点击“授权”。关闭页面。



单击“重新加载”



等待更新完毕即可。

2.2 设置sources.list

```
sudo sh -c 'echo "deb http://packages.ros.org/ros/ubuntu $(lsb_release -sc) main"
> /etc/apt/sources.list.d/ros-latest.list'
```

2.3 添加密钥

```
sudo apt-key adv --keyserver hkp://keyserver.ubuntu.com:80 --recv-key
C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654
```

```
uniwise@uniwise:~$ sudo apt-key adv --keyserver hkp://keyserver.ubuntu.com:80 --
recv-key C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654
Executing: /tmp/tmp.eeuwou9rna/gpg.1.sh --keyserver
hkp://keyserver.ubuntu.com:80
--recv-key
C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654
gpg: requesting key AB17C654 from hkp server keyserver.ubuntu.com
gpg: key AB17C654: public key "Open Robotics <info@osrfoundation.org>" imported
gpg: Total number processed: 1
gpg: imported: 1 (RSA: 1)
uniwise@uniwise:~$
```

2.4 更新package

```
sudo apt-get update
sudo apt-get upgrade
```

2.5 安装ROS kinetic完整版

```
sudo apt-get install ros-noetic-desktop-full
```

```
Setting up ros-noetic-rviz-plugin-tutorials (0.11.0-1focal.20221209.231034) ...
Setting up ros-noetic-rqt-reconfigure (0.5.5-1focal.20221209.205649) ...
Setting up ros-noetic-rqt-robot-dashboard (0.5.8-1focal.20221209.214118) ...
Setting up ros-noetic-visualization-tutorials (0.11.0-1focal.20221209.231532) ..
.
Setting up ros-noetic-ros-core (1.5.0-1focal.20221124.043914) ...
Setting up ros-noetic-geometry (1.13.2-1focal.20221209.221400) ...
Setting up ros-noetic-rqt-srv (0.4.9-1focal.20221209.214249) ...
Setting up ros-noetic-rqt-action (0.4.9-1focal.20221209.205757) ...
Setting up ros-noetic-urdf-sim-tutorial (0.5.1-1focal.20221209.223812) ...
Setting up ros-noetic-gazebo-plugins (2.9.2-1focal.20221209.220915) ...
Setting up ros-noetic-rqt-launch (0.4.9-1focal.20221209.205644) ...
Setting up ros-noetic-gazebo-ros-pkgs (2.9.2-1focal.20221209.230134) ...
Setting up ros-noetic-laser-pipeline (1.6.4-1focal.20221209.224944) ...
Setting up ros-noetic-ros-base (1.5.0-1focal.20221209.203123) ...
Setting up ros-noetic-rqt-robot-plugins (0.5.8-1focal.20221209.231519) ...
Setting up ros-noetic-robot (1.5.0-1focal.20221209.225949) ...
Setting up ros-noetic-rqt-common-plugins (0.4.9-1focal.20221209.214337) ...
Setting up ros-noetic-perception (1.5.0-1focal.20221209.225046) ...
Setting up ros-noetic-viz (1.5.0-1focal.20221209.231555) ...
Setting up ros-noetic-desktop (1.5.0-1focal.20221209.231659) ...
Setting up ros-noetic-simulators (1.5.0-1focal.20221209.231603) ...
Setting up ros-noetic-desktop-full (1.5.0-1focal.20221209.231810) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
```

2.6 安装辅助包

```
sudo apt install python3-rosdep2
```

2.7 初始化rosdep

```
sudo rosdep init
rosdep update
```

在初始化rosdep的时候，无法访问对应的域名，可以按照以下步骤进行：

1. 修改/usr/lib/python3/dist-packages/rosdep2/sources_list.py，加入代码

```
url="https://ghproxy.com/"+url

def download_rosdep_data(url):
    """
    :raises: :exc:`DownloadFailure` If data cannot be
            retrieved (e.g. 404, bad YAML format, server down).
    """
    try:
        url = 'https://ghproxy.com/' + url
        f = urlopen(url, timeout=DOWNLOAD_TIMEOUT)
        text = f.read()
        f.close()
        data = yaml.safe_load(text)
        if type(data) != dict:
            raise DownloadFailure('rosdep data from [%s] is not a YAML dictionary' % (url))
        return data
    except (URLError, httpplib.HTTPException) as e:
        raise DownloadFailure(str(e) + ' (%s)' % url)
    except yaml.YAMLError as e:
        raise DownloadFailure(str(e))
```

2. 将/usr/lib/python3/dist-packages/rosdistro/_init_.py里面的DEFAULT_INDEX_URL 进行如下替换：DEFAULT_INDEX_URL = <https://ghproxy.com/https://raw.githubusercontent.com/ros/rosdistro/master/index-v4.yaml>

```

# same version as in:
# - setup.py
# - stdeb.cfg
__version__ = '0.9.0'

# index information

#DEFAULT_INDEX_URL = 'https://raw.githubusercontent.com/ros/rosdistro/master/index-v4.yaml'
DEFAULT_INDEX_URL = 'https://ghproxy.com/https://raw.githubusercontent.com/ros/rosdistro/-
master/index-v4.yaml'

def get_index_url():
    # environment variable has precedence over configuration files
    if 'ROSDISTRO_INDEX_URL' in os.environ:
        return os.environ['ROSDISTRO_INDEX_URL']

    def read_cfg_index_url(fname):
        try:
            with open(fname) as f:
                return yaml.safe_load(f.read())['index_url']
        except (IOError, KeyError, yaml.YAMLError):
            return None

```

3. 在/usr/lib/python3/dist-packages/rosdep2/gbpdistro_support.py 第204行添加如下代码:

```

gbpdistro_url = "https://ghproxy.com/" + gbpdistro_url

def download_gbpdistro_as_rosdep_data(gbpdistro_url, targets_url=None):
    """
    Download gbpdistro file from web and convert format to rosdep distro data.

    DEPRECATED: see REP137. This function will output
    (at least) one deprecation warning

    :param gbpdistro_url: url of gbpdistro file, ``str``
    :param target_url: override URL of platform targets file
    :raises: :exc:`DownloadFailure`
    :raises: :exc:`InvalidData` If targets file does not pass cursory
    validation checks.
    """
    # we can convert a gbpdistro file into rosdep data by following a
    # couple rules
    # will output a warning
    targets_data = download_targets_data(targets_url=targets_url)
    try:
        gbpdistro_url = "https://ghproxy.com/" + gbpdistro_url
        f = urlopen(gbpdistro_url, timeout=DOWNLOAD_TIMEOUT)
        text = f.read()
        f.close()
        gbpdistro_data = yaml.safe_load(text)
        # will output a warning
        return gbprepo_to_rosdep_data(gbpdistro_data,
                                      targets_data,
                                      gbpdistro_url)

```

4. 在/usr/lib/python3/dist-packages/rosdep2/gbpdistro_support.py中加入
 FUERTE_GBPDISTRO_URL="https://ghproxy.com/" + FUERTE_GBPDISTRO_URL,
 DOWNLOAD_TIMEOUT = 15.0改成DOWNLOAD_TIMEOUT = 500.0

```

from .rosdistrohelper import get_targets, get_release_file, PreRep137Warning
from .rep3 import download_targets_data # deprecated, will output warning
import warnings

create_default_installer_context = None

# py3k
try:
    unicode
except NameError:
    basestring = unicode = str

# location of an example gbpdistro file for reference and testing

FUERTE_GBPDISTRO_URL = 'https://raw.githubusercontent.com/ros/rosdistro/' \
    'master/releases/fuerte.yaml'
FUERTE_GBPDISTRO_URL = "https://ghproxy.com/" + FUERTE_GBPDISTRO_URL

# seconds to wait before aborting download of gbpdistro data
DOWNLOAD_TIMEOUT = 500.0

```

5. 依照步骤4, 在/usr/lib/python3/dist-packages/rosdep2/sources_list.py中添加
 DEFAULT_SOURCES_LIST_URL="https://ghproxy.com/" + DEFAULT_SOURCES_LIST_URL,
 将DOWNLOAD_TIMEOUT = 15.0改成DOWNLOAD_TIMEOUT = 500.0

6. 依照步骤4, 在/usr/lib/python3/dist-packages/rosdep2/rep3.py中添加
EP3_TARGETS_URL="<https://ghproxy.com/>"+REP3_TARGETS_URL, 将
DOWNLOAD_TIMEOUT = 15.0改成DOWNLOAD_TIMEOUT = 500.0
7. 在/usr/lib/python3/dist-packages/rosdistro/manifest_provider/github.py中的第68行和
119行添加: url="<https://ghproxy.com/>"+url
8. 重新执行rosdep update

```
reading in sources list data from /etc/ros/rosdep/sources.list.d
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/osx-homebrew.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/base.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/python.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/ruby.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/releases/fuerte.yaml
Query rosdistro index https://ghproxy.com/https://raw.githubusercontent.com/ros/rosdis
tro/master/index-v4.yaml
Skip end-of-life distro "ardent"
Skip end-of-life distro "bouncy"
Skip end-of-life distro "crystal"
Skip end-of-life distro "dashing"
Skip end-of-life distro "eloquent"
Add distro "foxy"
Skip end-of-life distro "galactic"
Skip end-of-life distro "groovy"
Add distro "humble"
Skip end-of-life distro "hydro"
Skip end-of-life distro "indigo"
Skip end-of-life distro "jade"
Skip end-of-life distro "kinetic"
Skip end-of-life distro "lunar"
Add distro "melodic"
Add distro "noetic"
Add distro "rolling"
updated cache in /home/uniwise/.ros/rosdep/sources.cache
```

2.8 配置ROS环境

```
echo "source /opt/ros/noetic/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

2.9 安装rosinstall

```
sudo apt install python3-rosinstall python3-rosinstall-generator python3-wstool
```

2.10 测试ROS是否安装成功

2.10.1 CTRL+ALT+T打开终端, 输入以下命令, 初始化ROS环境:

```
roscore
```

```
uniwise@uniwise-virtual-machine:~$ roscore
... logging to /home/uniwise/.ros/log/e0623b42-b0c7-11ed-8399-072dcf2e3b75/roslaunch-uniwise-virtual-machine-100168.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://uniwise-virtual-machine:42709/
ros_comm version 1.15.15

SUMMARY
=====

PARAMETERS
* /roscdistro: noetic
* /rosversion: 1.15.15

NODES

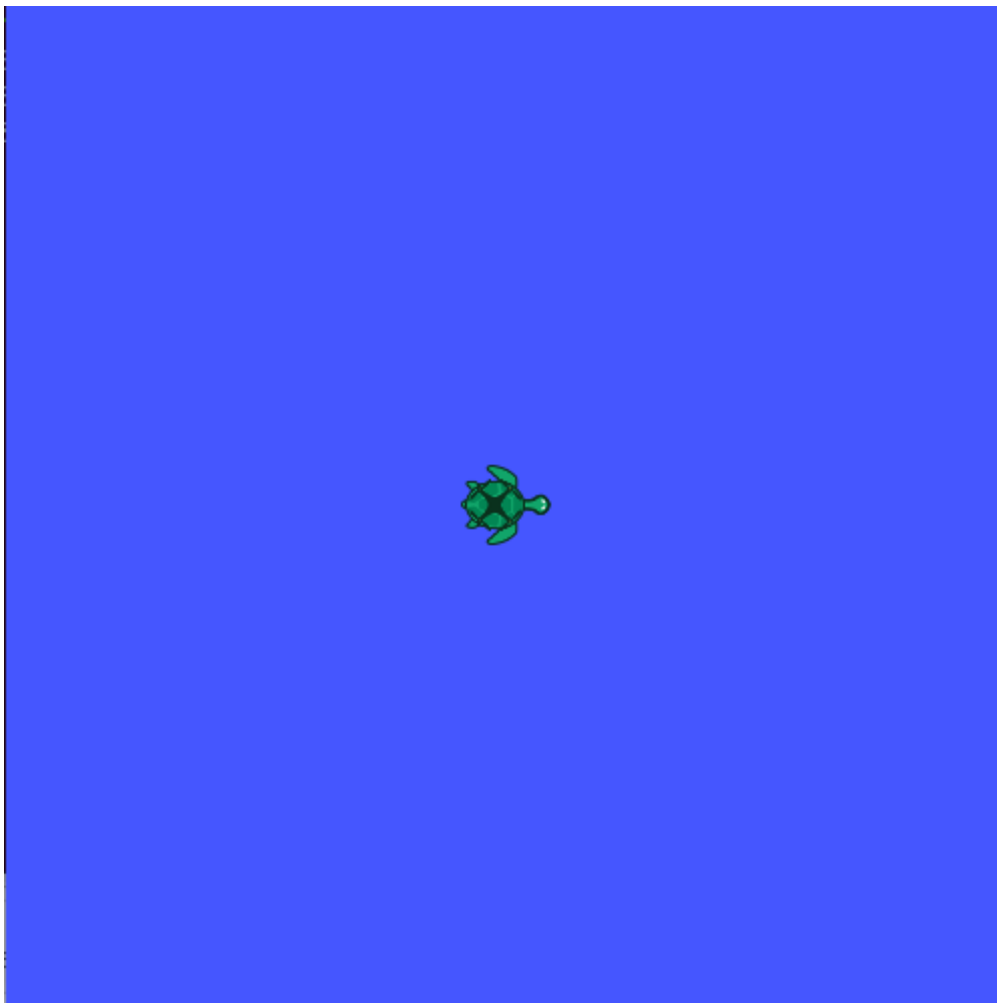
auto-starting new master
process[rosmaster]: started with pid [100176]
ROS_MASTER_URI=http://uniwise-virtual-machine:11311/

setting /run_id to e0623b42-b0c7-11ed-8399-072dcf2e3b75
process[roscout-1]: started with pid [100186]
started core service [/roscout]
```

以上界面表示ros安装成功。

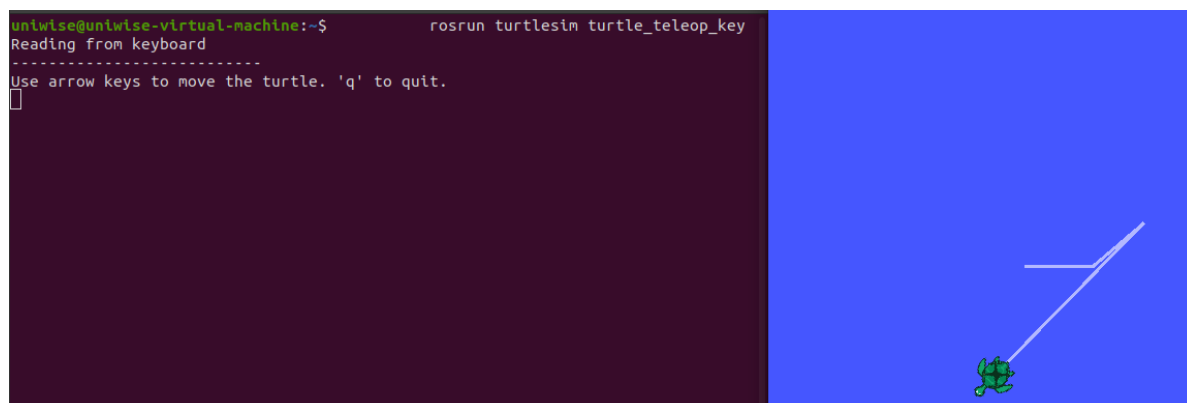
2.10.2 打开一个新的终端，输入以下命令，弹出一个小乌龟窗口：

```
roslaunch turtlesim turtlesim_node
```



2.10.3 打开一个新的终端，输入以下命令，用键盘方向键控制小乌龟移动

```
roslaunch turtlesim turtlesim_key
```

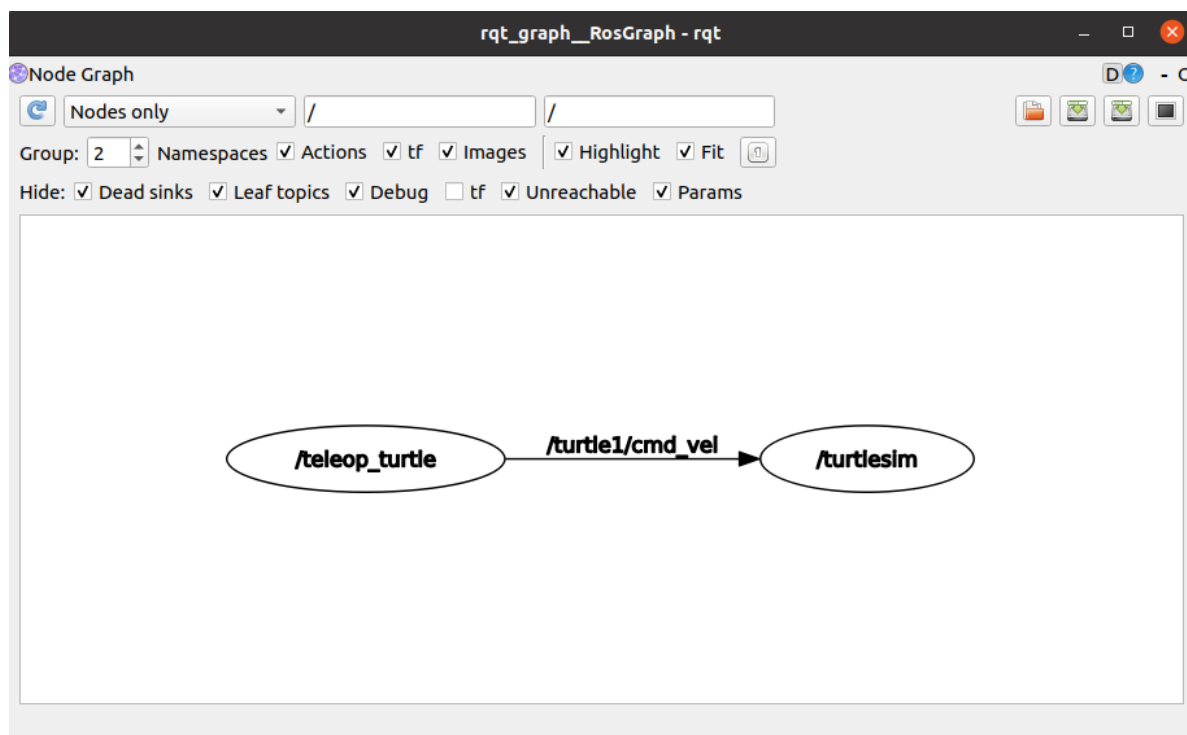


可以通过键盘方向键移动小乌龟。

注意：以上命令窗口和小乌龟窗口并列，激活命令窗口，按方向键，可以发现小乌龟在移动。如果激活的是小乌龟窗口，按方向键小乌龟是不动的。

2.10.4 打开新的Terminal，输入以下命令，可以查看ROS节点信息：

```
roslaunch rqt_graph rqt_graph
```



至此，ROS安装成功。

三 实验总结

1. ROS为Robot Operating System（机器人操作系统）的简写，是一个面向机器人的开源元操作系统（open-source, meta-operating system）。主要目标是为机器人研究和开发提供代码复用的支持。ROS也支持一种类似于代码储存库的联合系统，所有的工程都可以被ROS的基础工具整合在一起。ROS在高层架构上提供了更多的算法应用（如集成OpenCV的视觉算法）。

2. ROS的特点：

- 点对点设计
- 不依赖编程语言
- 精简与集成
- 便于测试
- 开源
- 强大的库和社区